Pedro Henrique Muralha Paranhos

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EDUCATION

Federal University of Rio de Janeiro, UFRJ — Rio de Janeiro, RJ
Bachelor's degree in Mechanical Engineering — Junho de 2021- December 2027 (projected)

EXPERIENCE

Project Intern — Rio de Janeiro, RJ
Moove (Cosan Lubricants and Specialties SA) — September 2024 – July 2025

- I implemented an official engineering project checklist, covering scope, schedule, technical documentation, approvals, and acceptance criteria, reducing wasted time, rework, and execution errors, and standardizing quality control of deliverables.
- Direct leadership in smaller-scale projects, from the quotation phase to installation, focusing on instrumentation and civil renovations.
- Technical support for highly complex projects, including the optimization of grease and oil mixers, pipeline automation with pneumatic valves, and civil engineering infrastructure.
- Support in weekly safety presentations and project status updates.
- Assistance in developing a Capex plan.
- Managing the purchasing process and initiating hiring procedures to ensure the physical and financial schedule of deliveries is met.
- Direct support to designers in the development of technical drawings in AutoCAD, ensuring fidelity between the designed scope and the execution in the field.

PROJECTS

Complex AI agents with Pythonhttps://github.com/pedromparanhos-sys/complex-ai-agents-langgraph-course

- I developed 5 AI agents in Python using LangGraph and LangChain (GPT-4o), implementing tool calling patterns (ReAct), conversational memory, and human-in-the-loop flows, orchestrated as state graphs with conditional routing.
- I built an end-to-end RAG (Retrieval-Augmented Generation) pipeline, integrating PyPDF, OpenAI embeddings, and ChromaDB as a vector store, to answer questions about documents with source citations.

Statistical Pairs Trading with Kalman Filter (Python)https://github.com/pedromparanhos-sys/pairs-trading

- I developed a complete pairs trading pipeline for S&P 500 stocks, implementing cointegration tests (Engle-Granger, Johansen) and a Kalman Filter for dynamic hedge ratios, recovering spread stationarity where static methods failed (ADF p-value 0.011 vs 0.63).
- I validated the strategy via walk-forward out-of-sample across 12 six-month windows (2018–2026), obtaining an average Sharpe ratio of 0.79 with 91.7% of windows showing positive results.
- I have honestly documented limitations (costs consuming 62% of the gross alpha) with a complete mathematical appendix of the derivations (cointegration, Kalman, Ornstein-Uhlenbeck process).

Predictive Microstructure Signal in Order Book (Python, Binance API)https://github.com/pedromparanhos-sys/orderbook-signal

- I built a real-time collection pipeline via WebSocket of 3M+ snapshots of the L2 order book (BTC/USDT), with feature engineering for Order Book Imbalance and Trade Flow Imbalance.
- I identified monotonic signal decay between prediction horizons via purged k-fold cross-validation (AUC 0.89 at 1s, 0.61 at 30s), replicating results from the microstructure literature (Cont, Stoikov & Talreja, 2010).
- I quantified a cost/signal ratio of 17:1 in a backtest with real transaction costs, concluding that the statistical edge exists but is unfeasible with taker execution, a conclusion validated with sensitivity analysis across 6 thresholds.

Modelo de Leveraged Buyout (LBO) - Fleury S.A. (B3: FLRY3)
https://github.com/pedromparanhos-sys/lbo-fleury

- I built a complete LBO model for Fleury S.A. (B3: FLRY3), integrating Excel and Python, with a projected return of 2.74x MOIC / 22.3% IRR on a capital structure of R$ 13.96 billion.
- I reconciled historical financial data directly from the CVM (Brazilian Securities and Exchange Commission), identifying leverage distortions between financial debt and lease liabilities (IFRS 16).
- I developed a Monte Carlo simulation engine in Python (10,000 scenarios, correlated variables), integrated with Excel via openpyxl, to quantify risk-return with 6 decimal place precision.

SKILLS

- **Software:** Microsoft Office, MS Project , Power BI, SolidWorks, Python (Anaconda, IPython, Jupyter Notebook, Pandas, NumPy, Matplotlib, SciPy, Seaborn, Scikit-Learn, Statsmodels, Polars, Plotly), LangChain, LangGraph, SQL, HTML, CSS, JS, Claude Code.
- **Languages:**Portuguese (native), English (advanced).